

Phase I/II Operating Characteristics

One-stage versus two-stage allocation strategies - N = 30, Cycle_Max = 2, efficacy futility cutoff = 0.85

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
AIDE one-stage	10.1	8.1	4.3	0.9	0.1	76.5
AIDE two-stage (highest utility)	10	9.8	3.9	0.8	0	75.5
AIDE two-stage (top-2 randomized)	12.1	7.3	2.8	1.1	0	76.7
Mean patients treated						
AIDE one-stage	11.7	4.3	1.7	0.7	0.2	
AIDE two-stage (highest utility)	11.6	4.4	1.8	0.7	0.2	
AIDE two-stage (top-2 randomized)	11.3	4.6	1.9	0.7	0.2	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
AIDE one-stage	28.6	23.4	19.6	17.6	10.8	0
AIDE two-stage (highest utility)	30.5	23.7	18	15.9	11.9	0
AIDE two-stage (top-2 randomized)	33	27.3	17.2	13.2	9.2	0.1
Mean patients treated						
AIDE one-stage	8	6.8	5.7	5.3	4.2	
AIDE two-stage (highest utility)	7.7	6.8	5.7	5.4	4.4	
AIDE two-stage (top-2 randomized)	7.7	6.8	5.6	5.5	4.4	

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
AIDE one-stage	10.2	49.6	23.3	10.9	2.6	3.4
AIDE two-stage (highest utility)	9.8	48.7	25.2	11.3	2.2	2.8
AIDE two-stage (top-2 randomized)	7.1	53.2	22.7	11	3.4	2.6
Mean patients treated						
AIDE one-stage	6.3	11	6.7	4.1	1.4	
AIDE two-stage (highest utility)	6	10.2	7.2	4.5	1.7	
AIDE two-stage (top-2 randomized)	7	9.5	7.2	4.3	1.6	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
AIDE one-stage	1.4	1.5	5.2	58.8	33.1	0
AIDE two-stage (highest utility)	1.5	1.6	6	55.7	35.2	0
AIDE two-stage (top-2 randomized)	0.8	1	4.4	56.9	36.9	0
Mean patients treated						
AIDE one-stage	3.5	3.5	4.4	12.3	6.3	
AIDE two-stage (highest utility)	3.5	3.5	4.2	10.7	8.2	
AIDE two-stage (top-2 randomized)	3.9	4	4.9	9.2	8	

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
AIDE one-stage	14.1	24	44	11	0.7	6.2
AIDE two-stage (highest utility)	15.1	22.4	44.8	10.6	1.3	5.8
AIDE two-stage (top-2 randomized)	11.6	24.2	44.8	12.7	1.1	5.6
Mean patients treated						
AIDE one-stage	7.9	7.5	9.4	3.6	0.6	
AIDE two-stage (highest utility)	7.4	7.4	9.3	4	1.1	
AIDE two-stage (top-2 randomized)	8.3	8.1	8	3.9	1.1	
Scenario 38						
DLT rate	0.05	0.1	0.12	0.15	0.3	
Efficacy rate	0.05	0.08	0.12	0.15	0.5	
Utility 3	41	40.8	42.4	43	58	
OBD selection (%)						
AIDE one-stage	8.9	10.9	9.1	9.5	60.4	1.2
AIDE two-stage (highest utility)	10.1	11	8.9	9	60.1	0.9
AIDE two-stage (top-2 randomized)	9.2	10.8	9.3	8.9	60.8	1
Mean patients treated						
AIDE one-stage	5.6	5	4.5	4.8	9.9	
AIDE two-stage (highest utility)	5.4	5	4.5	4.9	10	
AIDE two-stage (top-2 randomized)	6.5	6	5	5	7.4	